

Boonyakorn Tanrattanakorn

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Education

Chulalongkorn University, BE in Computer Engineering – Bangkok, Thailand Sept 2023 – present

- GPA: 3.91/4.00
- **Relevant Coursework:** Computer Programming (A), Introduction to Data Structures (A), Operating Systems and Programs (A), Database Systems (A), Algorithm Design (A), Hardware Synthesis Laboratory (A), Embedded System Laboratory (A), Software Engineering (A), Introduction to Robotics (A), Computer Networks I (A)

Experience

Embedded System Engineer Intern, Ivory Electronics Co. Ltd. – Thailand June 2026 – present

- Architected a multilayered Arduino firmware (driver → manager → app → UI) for a hydrogen electrolysis teaching instrument, acquiring hydrogen separation current, half-cell voltage, and total dissolved solids via **ADC**.
- Diagnosed a faulty voltage reference IC causing ADC drift; implemented a **median filter** to stabilise readings and restore measurement accuracy.
- Designed a 3D-printed instrument enclosure in **Onshape**, including sensor mounts and wiring layout.

Projects

Project CURSR-III Jan 2023

Chulalongkorn University High Altitude Research Club | Avionics Engineer

- Designed and developed a telemetry ground station using **ESP32** and **LoRa**.
- Designed and assembled 2-layer throughhole **PCB** for ground station using **KiCad**.
- Participated in the fly-off team at the **Spaceport America Cup 2024** in USA.

Project CURSR-IV Jan 2024

Chulalongkorn University High Altitude Research Club | Avionics Engineer

- Designed and assembled a 4-layer SMT flight computer **PCB** using **KiCad**.
- Developed embedded firmware on **STM32 (C/C++)** for sensor integration, telemetry, and flight control.
- Built a **LoRa**-based mesh network ground station using **ESP32** and MQTT for telemetry diversity.
- Modeled and simulated an active airbrake control system in **Simulink**.
- Participated in the fly-off team at the **International Rocket Engineering Competition 2025** in USA.

Project XSRT Jan 2025

Chulalongkorn University High Altitude Research Club | Electrical Engineer

- Developed an R&D web application using **Next.js** and **Tailwind CSS** for a hybrid propulsion test bench user interface.
- Deployed **Flask** backend on **Raspberry Pi 5** to control the test bench.

FPGA Real-Time Video Player Jan 2025

Academic Project | Verilog

- Implemented a real-time FPGA (Basys-3) video player in **Verilog** with VGA output.
- Developed an SPI-based SD card controller to stream 320x240 RGB frames into dual-port RAM.

Smoothed Particle Hydrodynamics Simulator

Jan 2025

Personal Project | Godot

- Built a real-time 2D fluid dynamics simulator in **Godot Engine** using the SPH computational method written in **GDScript**.
- Implemented GPU-accelerated physics calculations with **GLSL compute shaders** for density and pressure force computations.

Weather Station

Jan 2025

Embedded System Lab I (2110366), Chulalongkorn University | Embedded System Programmer

- Developed **STM32** firmware in **C** to interface with DHT11 temperature/humidity, LDR light, and water level sensors via ADC and a custom one-wire protocol.
- Designed a compact binary UART data packet protocol for STM32-**ESP32-S3** communication with checksum validation.
- Implemented **ESP32-S3** firmware to publish sensor data to a cloud **MQTT** broker (Eclipse Mosquitto) and control a servo motor rain gauge reset via **PWM**.

Technical Skills

Embedded Systems: Integrated Circuits, SMT, PCB Design & Assembly, STM32 & ESP32 Programming, FPGA Programming

Programming: Python, C, C#, C++, Java, Verilog

CAD: Fusion 360, KiCad

Languages: Thai: Native, English: IELTS 7.5, Japanese: JLPT N4